## 3.21 DRAWING AND DESIGN (449)

**Drawing and Design Paper**

**SECTION A** marks

*Answer **all** the questions in this section on the answer sheet provided*

a List **four** characteristics of a good technical drawing paper marks

b Given that paper size A is 210 x 297. Determine the sizes of the following paper sizes

i A

ii A marks

c State **two** precautions in handling a T-square marks

a List **six** computer programmes that can be used to produce a drawing marks

(b) Define the term "mock-up" and state its purpose in the design process. (2 marks)

Name the **three** groups of metals and give **one** example in each group

marks

a **Figure** is drawn to scale of

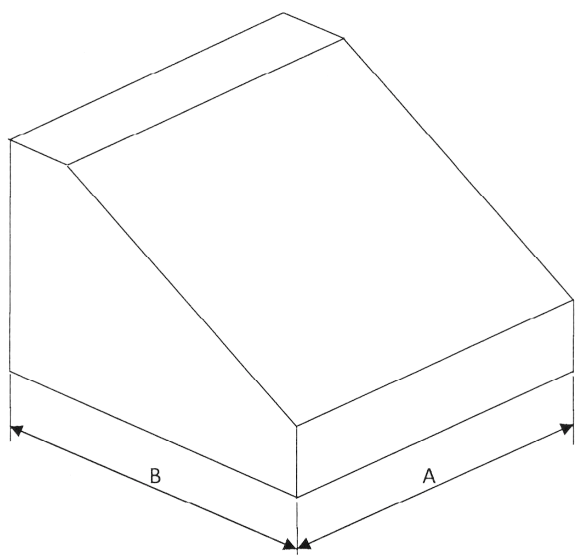



Figure 1

Determine:

i distance A

ii the angle of the slanting face

marks

b Sketch to show how the diameters of eccentric circles on a solid piece can be dimensioned marks

□ Define the following terms as applied to business enterprises:

(a) fixed assets;

(b) deficit;

□ □c□□ liability□□ □ □ □ □ □ □ □ □ □□□marks□

□□ **Figure□□**shows two views of two parts of a machine component drawn in first angle

□ projections□□Sketch□the□assembled□parts□in□oblique□projection□□ □ □ □□□marks□

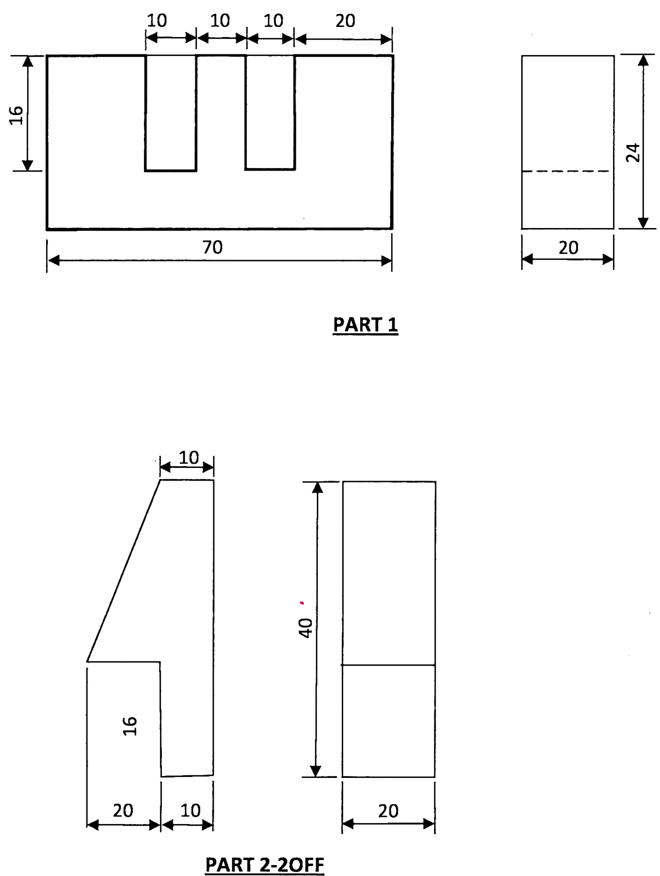



Figure 2

**Figure** shows the front elevation and an incomplete plan of a truncated square based pyramid.

(a) complete the plan.

(b) draw the true shape of the cut face.

marks

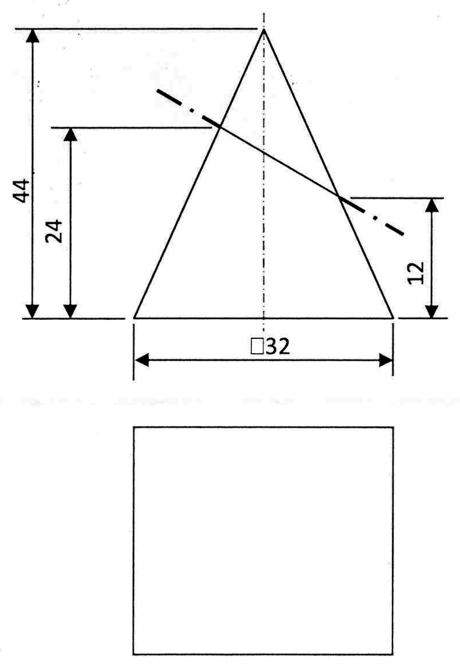



**Figure 3**

Draw the locus of the end of a string when it is unwound from a 30 mm square prism for one complete revolution. marks

**Figure** shows a block drawn in first angle projection. Sketch the block in oblique taking AB as the lowest edge. marks

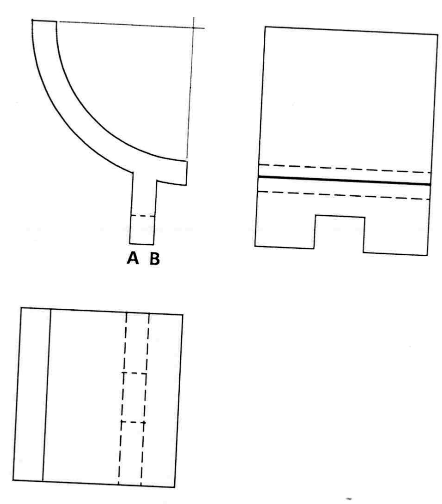



**Figure 4**

**Figure** shows an isometric block. Sketch three views of the block in first angle orthographic projection. marks

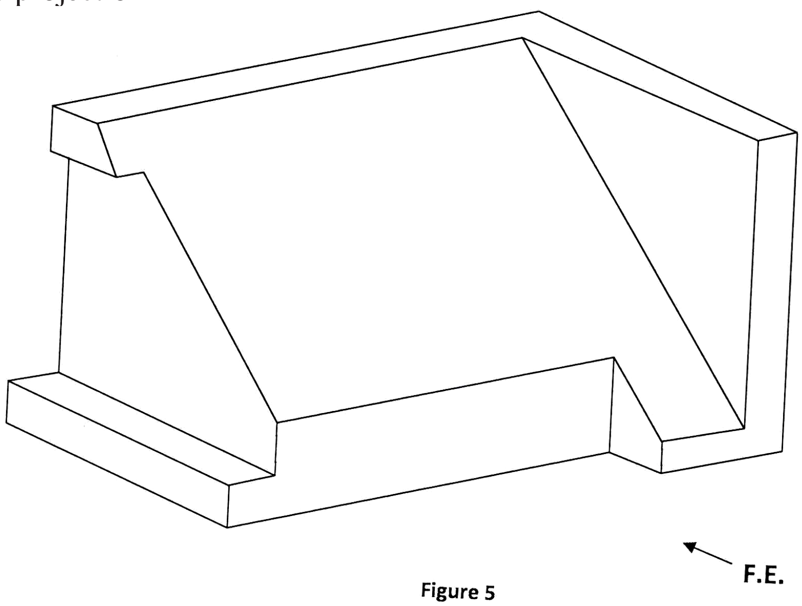



Figure 5

## SECTION B marks

*This question is* ***compulsory****. Candidates are advised to spend not more than one hour on this question.*

**Figure** shows parts of a mechanical component drawn in first angle projection. Assemble the parts and draw FULL SIZE the following:

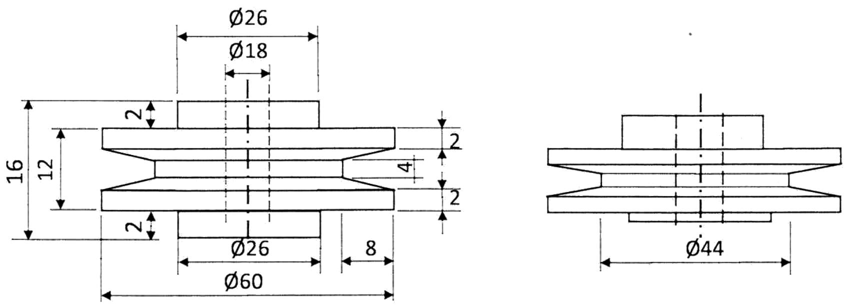
(a) sectional front elevation along the cutting plane P-P;

(b) end elevation;

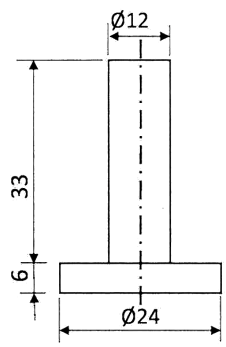
(c) insert three leading dimensions.

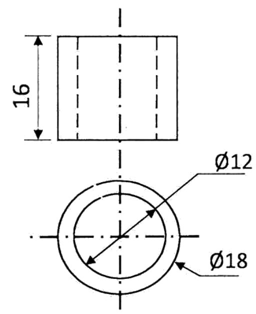
Unspecified dimensions are left to the candidates discretion. Hidden details are not required.

**Use the A paper provided.** marks



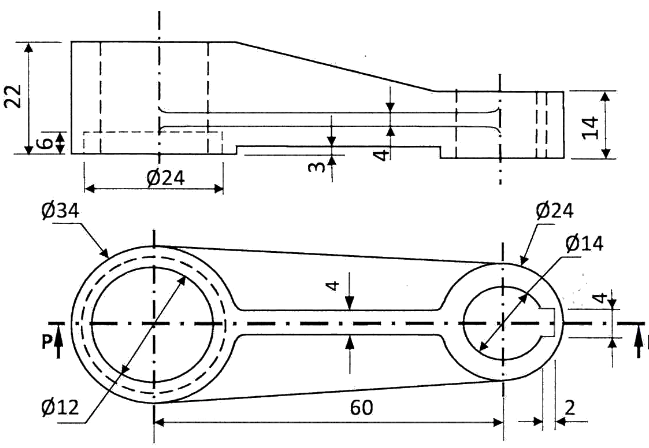

PART 1-PULLEY



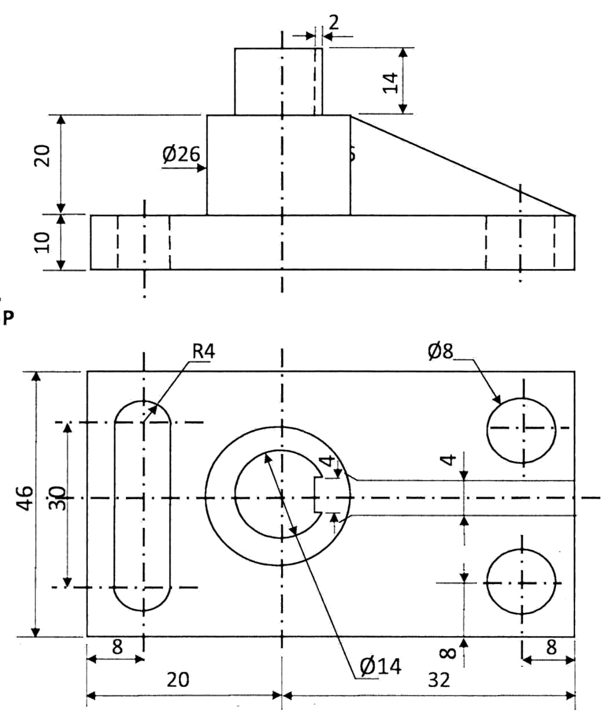



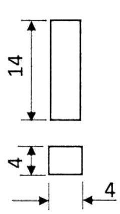

PART2 –PIN

PART 3- BUSH




PART 4 -CONNECTOR




PART 6 –KEY

PART 5 -BASE

Figure 6

**SECTION C** marks

*Answer any* ***two*** *questions from this section*

**Figure** shows the front elevation and an incomplete plan of a truncated hexagonal prism

(a) copy the views and complete the plan

(b) draw the surface development of the prism (omit the flaps).

marks

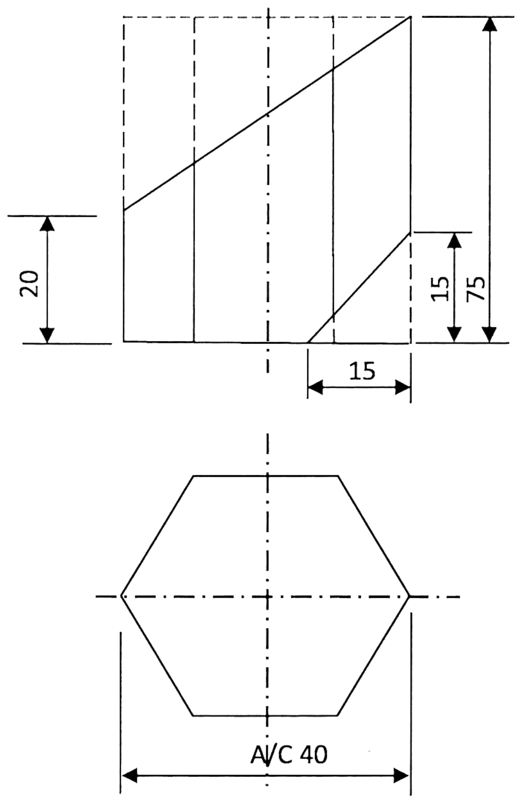



**Figure 7**

**Figure** shows an inclined plan of a block and its front elevation.

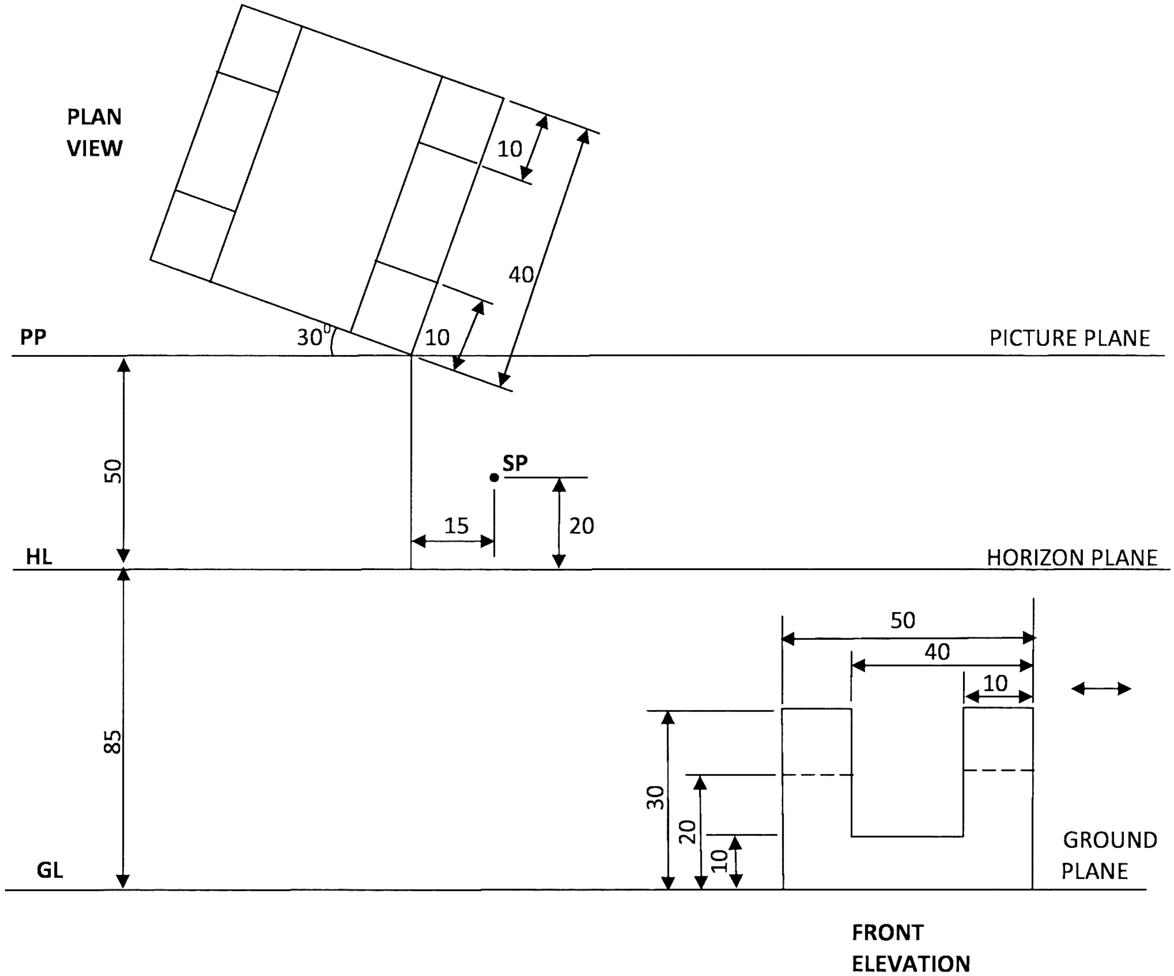



Figure 8

Copy the given layout and draw the two point perspective of the block showing all construction details. (marks)

**Figure 9** shows two intersecting square tubes A and B drawn in 1st angle projection.

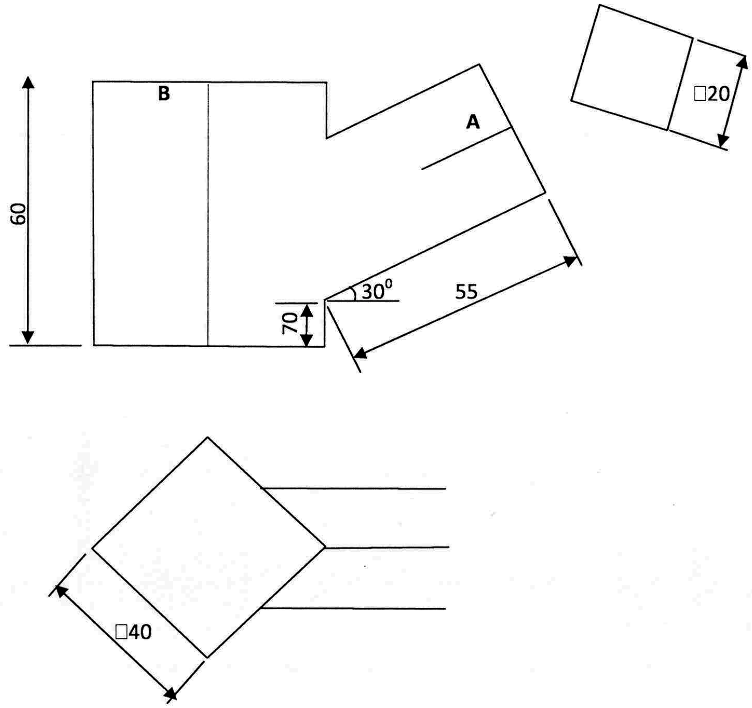



Figure 9

(a) copy the figure and complete:

(i) the front elevation

(ii) the plan;

(b) Draw the development of tube B.

(marks)